

Workforce ROI Calculator background at a glance

What it measures, what the numbers rest on, and why it is built this way. For the team that owns the tool.

— WHAT IT MEASURES

A Agency premium displaced premium you stop paying when a shift moves agency → bank	Cash the headline
B Admin time released roster / bank time freed by self-booking	Cash by default (toggle off)
C Capacity extra duties your bank now fills	Never cash shown apart

— THE CORE PRINCIPLE

Filling a bank shift still costs money. You save only the **premium** (agency price minus bank price), not the whole shift.

$$\text{cash} = \text{agency spend} \times \text{displacement} \times \text{premium} / (1 + \text{premium})$$

— WHAT THE NUMBERS REST ON

- **Pay:** 2026/27 NHS AfC midpoints, 1,957.5 h/yr, 8h shifts, 20% on-cost
- **Agency premium:** ~20% (House of Commons Library), contested, editable
- **Displacement stance:** Conservative ~13% (default), Pilot ~26%, Optimistic ~35%
- **Platform cost:** £17k/yr, the ROI denominator
- **Pilot evidence:** 2-site data in a 4-trust programme; indicative, not solely attributable

— TWO MODES, BY AUDIENCE

- Quick estimate:** three inputs (bank pool, agency fill, roster team). For operational managers and events.
- Detailed / commercial:** agency spend by staff group, editable premium and levers. For finance leads and workforce directors.

— CONFIGURABLE & PERSONALISED

Live, no code: pick an organisation type, set its size (total bank headcount and agency spend), then refine per staff group, premium, stance and optional levers. Everything recomputes instantly and is most credible on the trust's own rates.

In code (owner): one central constants block to re-base pay each year, rebrand, re-localise (en-GB, AU-ready), edit the org templates, and wire up lead capture.

— WHY IT IS BUILT THIS WAY

Defensible over impressive: conservative defaults, a transparent formula, every assumption shown with its source, and honest caveats. The same tool scales from a 30-second event estimate to a finance-grade business case.